Dexin Ren

M.S. Computer Science | Data-Centric & Reliable Machine Learning

Seeking Fall 2027 PhD Opportunities in Machine Learning

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Research Focus

Data-centric and reliable machine learning; robustness under distribution shift; model failure analysis; how learning objectives, representations, and data characteristics affect generalization.

Education

University of Southern California Jan 2025 – Dec 2026

M.S. Computer Science (General), GPA: 3.7/4.0

University of Illinois Urbana-Champaign Aug 2021 – May 2024

B.S. Mathematics + Computer Science, GPA: 3.62/4.0, High Distinction

Publications

Dexin Ren, Shang-Ling Hsu, Chen Chu, Lifeng Lin, Ryosuke Shibasaki, Noboru Koshizuka, and Cyrus Shahabi. *RectifiedTraj: Map-Free Causal GPS Trajectory Denoising with Rectified Flow.* **ACM SIGSPATIAL '26**, short paper (accepted; to appear).
doi.org/10.1145/3841645.3843430

Dexin Ren, Shang-Ling Hsu, Chen Chu, Lifeng Lin, Ryosuke Shibasaki, Noboru Koshizuka, and Cyrus Shahabi. *RectifiedTraj Revisited: An Expanded Evaluation of Map-Free Causal GPS Trajectory Denoising.* **UMFM '26** (ACM SIGSPATIAL workshop), accepted; to appear.
doi.org/10.1145/3849729.3856129

Hongyuan Dong, Jiaqing Xie, Zhi Jing, and **Dexin Ren**.
Variational Autoencoder for Anti-Cancer Drug Response Prediction. AI4PH Workshop at ICLR 2021.
arXiv:2008.09763

Research Experience

RectifiedTraj: Causal GPS Trajectory Denoising Aug 2025 – Sep 2026

Researcher / First-Author Lead, USC InfoLab

- Led first-author research from problem formulation through model and data-pipeline design, experiments, failure analysis, and manuscript preparation.
- Identified a failure of direct-coordinate prediction as trajectory scale increased; controlled comparisons showed that predicting corrections maintained stability across changing input conditions.
- Designed robustness evaluations across datasets, geography, simulators, noise conditions, and sampling intervals to test generalization beyond training data.
- Analyzed errors near the beginning of causal prediction windows and evaluated a method to reduce them, quantifying the accuracy–throughput trade-off.
- Improved real-world BlogWatcher accuracy from **44.63% to 57.05%** (**+12.42 percentage points**); demonstrated CPU-only mobile inference, with the hybrid model taking approximately **124 ms** per 256-point chunk on a Pixel 4a.

Research Experience (continued)

HAYSTAC: Human Mobility Anomaly Detection Jun 2026 – Sep 2026

Research Team Member, USC InfoLab & Novateur Research Solutions

- Investigated why a specialized mobility-anomaly Transformer underperformed a simpler MLP baseline, using controlled model comparisons to isolate contributing factors.
- Found that architecture explained only part of the performance gap; extended the investigation to input representations, training objectives, and synthetic supervision.
- Identified a model-selection problem: training and validation losses based on synthetic anomalies could be poorly aligned with performance on evaluation anomalies.
- Evaluated how training-loss rankings corresponded to detection performance across checkpoints; asymmetric focal loss showed the strongest agreement in the tested configurations.
- Redesigned the training data pipeline, approximately halving memory consumption and improving throughput by **2**–**12**× depending on configuration.

NeurogenesisGNN: EEG-Based Neurogenesis Estimation Jun 2026 – Present

Part-Time Research Assistant, USC IMSC & Bonaguidi Lab

- Built the end-to-end machine-learning pipeline for estimating patient-level hippocampal neurogenesis from EEG and clinical variables, adapting an existing representation model to the task.
- Recognized that approximately **13,000 EEG clips** represented only **30 independent patients**; addressed the resulting high-dimensional, small-sample learning problem with statistical modeling.
- Compared prediction methods and found that supervised dimensionality reduction with regularized regression performed best in the tested settings; distinguished clip-level development evaluation from unseen-patient generalization.
- Developed a training-derived percentile aggregation method to combine clip-level predictions into patient-level estimates.
- Investigated the difficult zero-neurogenesis subgroup using two-stage models and patient-level analyses; feature ablations and SHAP highlighted disease duration as an important predictive variable in the cohort.

TrafficWhatIf / DeepCitySim: Intervention-Aware Traffic Prediction Sep 2026 – Present

Team Lead / Researcher, USC InfoLab

- Leading ongoing research on intervention-aware traffic prediction, focusing on the data and evaluation requirements for learning from road-network changes.
- Developing a benchmark and evaluation strategy aimed at generalization to unseen cities and synthetic-to-real transfer; results are in progress.

Technical Skills

Languages: Python, C/C++, Java, SQL

ML & Data: PyTorch, scikit-learn, NumPy, SciPy, pandas

Systems: Linux, Docker, Git, CUDA